

MEDIATEK

MT8676 Android SDCard User Manual

Version: 1.0
Release date: 2024-08-12

Use of this document and any information contained therein is subject to the terms and conditions set forth in Exhibit 1. This document is subject to change without notice.

Version History

Version	Date	Author	Description
1.0	2024-08-12	Andy-Id Lu	Official release

Table of Contents

Version History	2
Table of Contents	3
List of Figures	3
List of Tables	3
1 SDCard	4
1.1 Overview	4
1.1.1 Brief Introduction	4
1.1.2 Abbreviations	4
1.2 Architecture/Process Overview	4
1.2.1 SDCard Introduction	4
1.2.2 MT8676 SDCard Feature	5
1.3 Configuration/Customization Guideline	5
1.3.1 Kernel Config	5
1.3.2 DTS Node	6
1.4 Frequently Asked Questions/Troubleshooting	7
1.4.1 SDCard is not Recognized, VDD Voltage could not be Measured	7
1.4.2 SDCard is Recognized while Power On with Card but Failed while Hot-plug	7
Exhibit 1 Terms and Conditions	8

List of Figures

Figure 1-1. Initialization flow of UHS-I card	5
Figure 1-2. DTS node for SDCard	6
Figure 1-3. Pinctrl node of SDR104 mode	6
Figure 1-4. dws setting for detect pin	7

List of Tables

Table 1-1. Abbreviations	4
--------------------------------	---

1 SDCard

1.1 Overview

1.1.1 Brief Introduction

This chapter introduces the hardware features, software configuration and function of SDCard controller, also contains the debugging methods for common problems.

1.1.2 Abbreviations

Table 1-1. Abbreviations

Abbreviation	Explanation
DDR50	Double Data Rate up to 50MB/s@50MHz
MMC	MultiMedia Card
SDR104	Signal Data Rate up to 104MB/s@208MHz
SDR12	Signal Data Rate up to 12.5MB/s@25MHz
SDR25	Signal Data Rate up to 25MB/s@50MHz
SDR50	Signal Data Rate up to 50MB/s@100MHz
UHS-I	Ultra High Speed Phase I card

1.2 Architecture/Process Overview

1.2.1 SDCard Introduction

SDCard is a new generation of high-speed storage device based on semiconductor flash memory, and developed from MultiMedia Card (MMC) format. It has high memory capacity, fast data transfer rate, great mobility flexibility, and good security, which is widely used on portable devices. In the SD3.0 protocol, theoretical maximum capacity of SDCard could be up to 2TB, the theoretical maximum speed of read/write could be up to 104MB/s.

The major pins and functions are described as below:

- (1) CLK: Clock signal, host controller or SDCard transmit one cmd/data bit at each clock cycle, up to 208MHz in UHS-I speed mode;
- (2) CMD: Command and response shared pin, command is transmitted from host controller to SDCard, and response is from SDCard to host controller;
- (3) DAT0~3: Data lines, data could be transmitted from SDCard to host controller(read), and could also be transmitted from host controller to SDCard (write);
- (4) VDD: Power supply pin of SDCard, which is normally configured to 3.3V voltage, the scope specified in the SD protocol is 2.7V~3.6V;
- (5) CD: Detect pin of SDCard, normally achieve GPIO level variation pin in the state of card inserting and removing through the mechanical structure of SDCard holder.

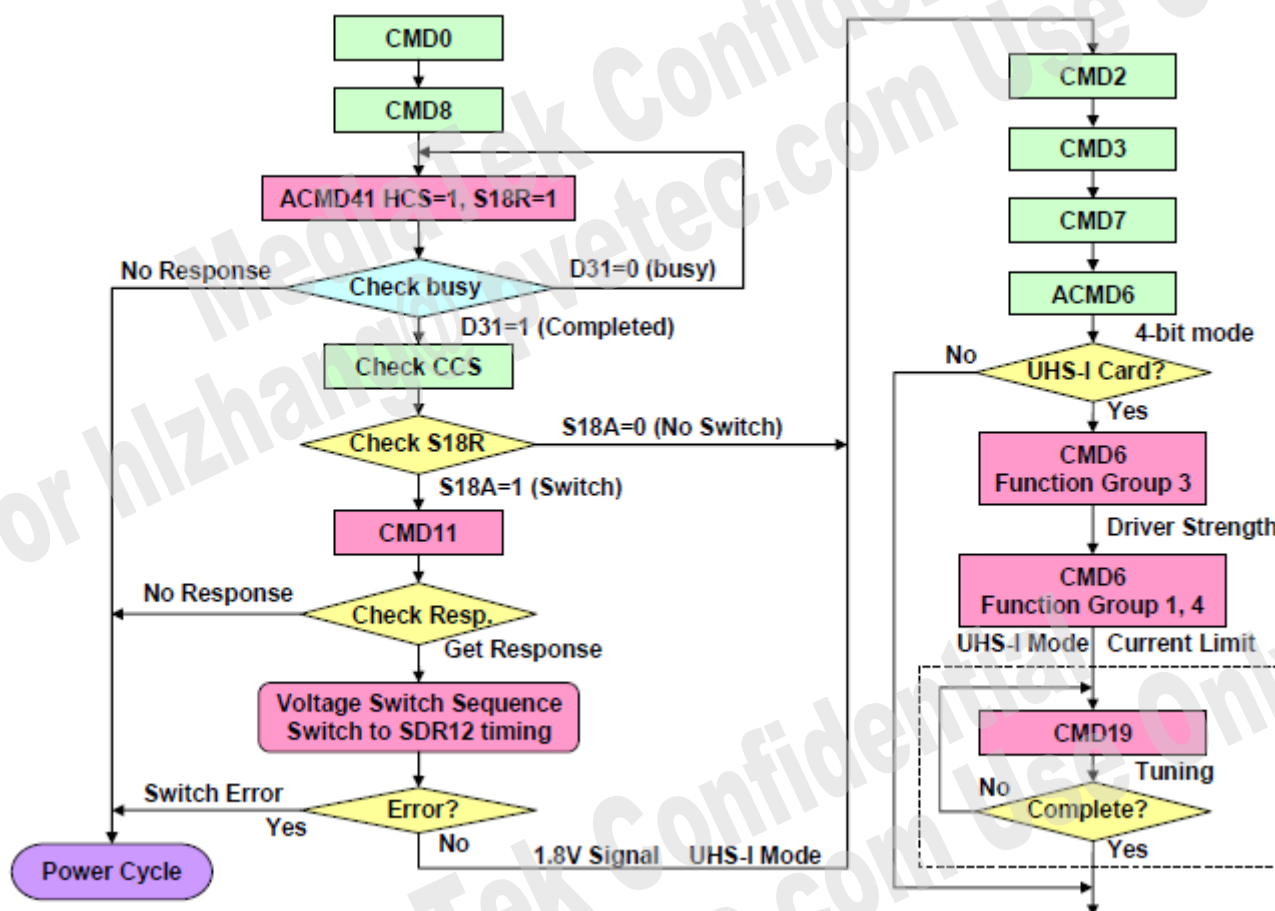


Figure 1-1. Initialization flow of UHS-I card

1.2.2 MT8676 SDCard Feature

- (1) Compatible with SD3.0 protocol standards
- (2) Supports Basic DMA and Descriptor DMA mode
- (3) Supports Bus speed mode: Default Speed/High Speed/SDR12/SDR25/SDR50/SDR104/DDR50
- (4) Supports 1/4bits bus width

1.3 Configuration/Customization Guideline

1.3.1 Kernel Config

- (1) Enables SDCard support
CONFIG_MMC = y
- (2) Enables Mediatek host driver support
CONFIG_MMC_MTK_PRO = m

1.3.2 DTS Node

```

10 &mmc1 {
11     host-index = <1>;
12     status = "okay";
13     pinctrl-names = "default", "state_uhs";
14     pinctrl-0 = <&mmc1_pins_default>;
15     pinctrl-1 = <&mmc1_pins_uhs>;
16     bus-width = <4>;
17     max-frequency = <200000000>;
18     ocr-voltage = <0x00070000>;
19     cap-sd-highspeed;
20     sd-uhs-sdr12;
21     sd-uhs-sdr25;
22     sd-uhs-sdr50;
23     sd-uhs-sdr104;
24     sd-uhs-ddr50;
25     cd-debounce-delay-ms = <0>;
26     cd-gpios = <&pio 33 GPIO_ACTIVE_LOW>;
27     vmmc-supply = <&mt6373_vmch_eint_high>;
28     vqmmc-supply = <&mt6373_vmc>;
29     no-mmc;
30     no-sdio;
31 };

```

Figure 1-2. DTS node for SDCard

- (1) SD2.0 card needs to configure "cap-sd-highspeed", SD3.0 ultra high speed card needs to configure "sd-uhs-xxx"
- (2) SD driving strength could be configured in the pinctrl node of the corresponding mode, such as the SDR104 mode shown as Figure 1-3;

```

74 mmc1_pins_uhs: mmc1@0{
75     pins-cmd-dat {
76         pinmux = <PINMUX_GPIO74_FUNC_MSDC1_DAT0>,
77                 <PINMUX_GPIO75_FUNC_MSDC1_DAT1>,
78                 <PINMUX_GPIO76_FUNC_MSDC1_DAT2>,
79                 <PINMUX_GPIO77_FUNC_MSDC1_DAT3>,
80                 <PINMUX_GPIO73_FUNC_MSDC1_CMD>;
81         input-enable;
82         drive-strength = <3>;
83         bias-pull-up = <MTK_PUPD_SET_R1R0_01>;
84     };
85
86     pins-clk {
87         pinmux = <PINMUX_GPIO72_FUNC_MSDC1_CLK>;
88         drive-strength = <3>;
89         bias-pull-down = <MTK_PUPD_SET_R1R0_10>;
90     };
91 };

```

Figure 1-3. Pinctrl node of SDR104 mode

- (3) SDCard detect pin could be configured by "cd-gpios", GPIO_ACTIVE_LOW represents low card insertion level, and GPIO_ACTIVE_HIGH represents high card insertion level
- (4) The "vmmc-supply" and "vqmmc-supply" are configured according to actual used SDCard VDD power and Host I/O power. If it is necessary to use fast-power-off (VMCH hardware power off when SDCard is pulled out) function, "vmmc-supply" should be configured as &mt6373_vmch_eint_high (match with GPIO_ACTIVE_LOW of "cd-gpios") or &mt6373_vmch_low (match with GPIO_ACTIVE_HIGH of "cd-gpios"); and if it is unnecessary to use fast-power-off function, "vmmc-supply" should be configured as &mt6373_vmch

1.4 Frequently Asked Questions/Troubleshooting

1.4.1 SDCard is not Recognized, VDD Voltage could not be Measured

- (1) Checks the correction of Kernel config and DTS according to the previous section;
- (2) If VDD supplied power is MT6373, and detect pin is connected to SD_DET pin of MT6373, check if the configured power node of "vmmc-supply" is matched with the polarity of detect pin;
- (3) If the result of step (2) is okay, configure "vmmc-supply" as &mt6373_vmch and check VDD voltage, correct voltage represents there is a problem in fast-power-off function, submit PMIC issue to MTK;
- (4) If VDD could not power up yet in step (3), catch kernel log and submit SDCard issue to MTK.

1.4.2 SDCard is Recognized while Power On with Card but Failed while Hot-plug

- (1) Checks the correction of "cd-gpios" configuration in DTS according to the previous section;
- (2) If DTS configuration is correct, check the correction of the GPIO configuration for detect pin in the file whose path is vendor/mediatek/proprietary/tools/dct/dws/mt6897/\${PROJECT}.dws, refer to the configuration shown in Figure 1-4;

```

430 <gpio33>
431     <eint_mode>true</eint_mode>
432     <def_mode>0</def_mode>
433     <inpull_en>true</inpull_en>
434     <inpull_selhigh>true</inpull_selhigh>
435     <def_dir>IN</def_dir>
436     <out_high>>false</out_high>
437     <varName0>GPIO_SDHC_EINT_PIN</varName0>
438     <smt>true</smt>
439     <ies>true</ies>
440 </gpio33>

```

Figure 1-4. dws setting for detect pin

- (3) If the DTS and dws configuration are all correct but SDCard could not be recognized yet while hot-plug, measure the level of detect pin in the state of card inserting and removing, submit SDCard issue to MTK if the result of measurement meets expectations.

Exhibit 1 Terms and Conditions

Your access to and use of this document and the information contained herein (collectively this “Document”) is subject to your (including the corporation or other legal entity you represent, collectively “You”) acceptance of the terms and conditions set forth below (“T&C”). By using, accessing or downloading this Document, You are accepting the T&C and agree to be bound by the T&C. If You don’t agree to the T&C, You may not use this Document and shall immediately destroy any copy thereof.

This Document contains information that is confidential and proprietary to MediaTek Inc. and/or its affiliates (collectively “MediaTek”) or its licensors and is provided solely for Your internal use with MediaTek’s chipset(s) described in this Document and shall not be used for any other purposes (including but not limited to identifying or providing evidence to support any potential patent infringement claim against MediaTek or any of MediaTek’s suppliers and/or direct or indirect customers). Unauthorized use or disclosure of the information contained herein is prohibited. You agree to indemnify MediaTek for any loss or damages suffered by MediaTek for Your unauthorized use or disclosure of this Document, in whole or in part.

MediaTek and its licensors retain titles and all ownership rights in and to this Document and no license (express or implied, by estoppels or otherwise) to any intellectual propriety rights is granted hereunder. This Document is subject to change without further notification. MediaTek does not assume any responsibility arising out of or in connection with any use of, or reliance on, this Document, and specifically disclaims any and all liability, including, without limitation, consequential or incidental damages.

THIS DOCUMENT AND ANY OTHER MATERIALS OR TECHNICAL SUPPORT PROVIDED BY MEDIATEK IN CONNECTION WITH THIS DOCUMENT, IF ANY, ARE PROVIDED “AS IS” WITHOUT WARRANTY OF ANY KIND, WHETHER EXPRESS, IMPLIED, STATUTORY, OR OTHERWISE. MEDIATEK SPECIFICALLY DISCLAIMS ALL WARRANTIES OF MERCHANTABILITY, NON-INFRINGEMENT, FITNESS FOR A PARTICULAR PURPOSE, COMPLETENESS OR ACCURACY AND ALL WARRANTIES ARISING OUT OF TRADE USAGE OR OUT OF A COURSE OF DEALING OR COURSE OF PERFORMANCE. MEDIATEK SHALL NOT BE RESPONSIBLE FOR ANY MEDIATEK DELIVERABLES MADE TO MEET YOUR SPECIFICATIONS OR TO CONFORM TO A PARTICULAR STANDARD OR OPEN FORUM.

Without limiting the generality of the foregoing, MediaTek makes no warranty, representation or guarantee regarding the suitability of its products for any particular purpose, nor does MediaTek assume any liability arising out of the application or use of any product, circuit or software. You agree that You are solely responsible for the designing, validating and testing Your product incorporating MediaTek’s product and ensure such product meets applicable standards and any safety, security or other requirements.

The above T&C and all acts in connection with the T&C or this Document shall be governed, construed and interpreted in accordance with the laws of Taiwan, without giving effect to the principles of conflicts of law.